5 A *cyclotron* is a device that accelerates charged particles to high energies.

The cyclotron uses a magnetic field to keep charged particles moving in circles. The charged particles move within two D-shaped hollow metal conductors called *dees*, which are separated by a very narrow gap.

An alternating square-wave potential difference is applied between the dees, resulting in a region of electric field at the gap. Each time the charged particles pass into the gap between the dees, they are accelerated by the potential difference across the gap. This increase their speed and also increases the radius of curvature of their path.

After many revolutions, the charged particles acquire high kinetic energies and exit the outer edge of the cyclotron.

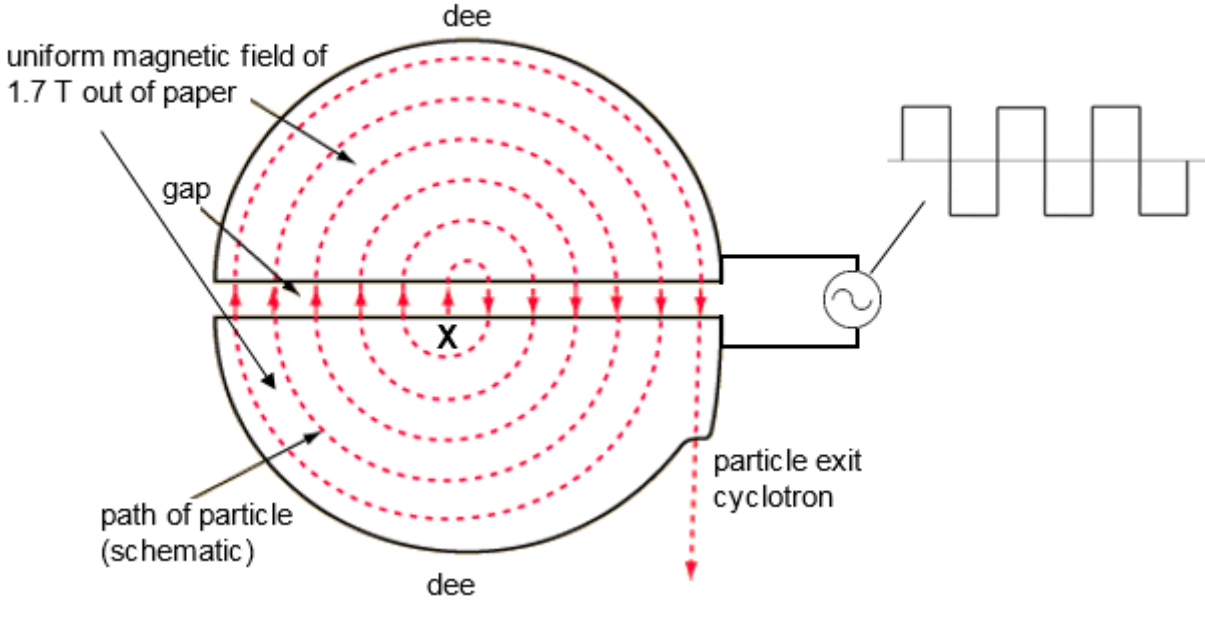

Fig. 5.1

A particle, enters the centre **X** of the cyclotron, as shown in Fig. 5.1, at a *low speed*, and travels in circular paths under the influence of a uniform magnetic field of flux density 1.7 T, which is acting perpendicularly out the page.

(a) The particle is either an electron or a proton. With reference to Fig 5.1, state what the particle is supposed to be.

..... [1]

(b) **(i)** The frequency of the alternating potential difference applied between the dees has to be set accurately to be equal to the frequency of revolutions of the charged particle.

Explain what will happen if the frequency is not set accurately.

.....

..... [1]

(ii) Show that the frequency needed for the alternating potential difference is 2.6×10^7 Hz.

[3]

(c) Just before exiting the cyclotron, the particle was traveling in a path of radius 0.25 m and had just completed 100 revolutions.

Determine

(i) the kinetic energy of the particle just before it exits the cyclotron.

kinetic energy = …… J [3]

(ii) the amplitude of the alternating potential difference applied across the dees.

amplitude = …… V [2]